Project Overview

Dolaptan · MTM4681 Mobile Programming · Milestone · Team [TAKIM NO] · [AD SOYAD 1], [AD SOYAD 2]

1. Problem and user

Our users are students and small households in Türkiye who cook at home. Our starting assumption is that they open the fridge, see ingredients, and still cannot decide what to make, which can lead to a takeaway order or food that spoils. The interviews below test this assumption and show how often it happens.

2. Evidence

Interviews: [TO FILL: interview 5-10 people who cook at home. Write how many people you spoke to, how many struggle to decide what to cook, how often this happens each week, and one short quote. Do not submit this placeholder.]

Statistic: The UNEP Food Waste Index Report 2024 estimates 1.05 billion tonnes of food waste worldwide in 2022, and around 60% of it came from households. Its annex lists a Türkiye household estimate of 102 kg per person per year (about 8.7 million tonnes). This is a **low-confidence** value: the report states it found no household study for Türkiye, so the number is extrapolated from other countries. We use it as context, not as proof of our problem.

3. Why existing apps fall short

App	What we verified (2 Nov 2026)	Gap for our user
Nefis Yemek Tarifleri	Strong Turkish recipe brand: 4.8 from 94k ratings on the Turkish App Store. Its website offers ingredient selection and an AI assistant. The app listing says it does not work without internet	Listing does not mention photo detection. No offline use. Shared list [TO CHECK]
SnapChef AI	Detects ingredients from a fridge or pantry photo. Premium costs $14.99 per month or $19.99-$29.99 per year	Turkish recipes and interface [TO CHECK]
Recipez	Fridge scan and 3 recipes, like ours. Filters listed: Italian, Chinese, Thai, Mexican, Indian. Exports to Walmart. 4 ratings	No listed Turkish focus. US grocery link. Paid plans after trial
Samsung Food	Phone photo scanning is reported as part of paid Food+	Scan reported as paid only. Turkish support [TO CHECK]

Photo-to-recipe apps and Turkish recipe apps both exist. We did not confirm any app that combines all four parts: Turkish-first recipes, editable photo detection, offline saved recipes, and a shared household list. This is a desk check of store listings, not a full market review. **[TO CHECK: install the apps above and test them in Turkish before submission.]**

4. Must-have features and course topics

Must-have feature	Course topic
Photo to editable ingredient list and 3 recipes (firebase_ai + App Check)	HTTP/API, JSON, async (Future)
Saved recipes that open without internet	Local DB (Hive)
Sign in, and a real-time shared shopping list	Firebase Auth, Firestore, Streams, StreamBuilder
Multi-step flow from photo to recipe, with bottom tabs	Navigation, go_router
Scan animation and recipe transitions	Animations and gestures
Separation of screens, state and data	MVVM, Provider

5. Weekly plan

Course week (topic)	[AD SOYAD 1]	[AD SOYAD 2]
7 (milestone)	Interviews, Figma import, prototype links	Review PDFs, GitHub repository
8 (midterm)	Flutter project, go_router routes, screen skeletons	Firebase project, Auth, App Check debug setup
9 (Hive)	Saved recipes in Hive, Offline status	Camera, AI ingredient detection, editable list
10 (Firestore, Auth)	Household, shopping list stream, security rules	Recipe generation with JSON schema, recipe screens
11 (Animations)	Scan animation, recipe transitions	Error states, retry, response validation
12 (Device, publishing)	Real-phone tests with two accounts, bug fixes	APK build, README, screenshots
13 (Presentations)	5-minute demo and presentation	2-minute backup video, peer contribution form

6. Risks and responses

Risk	Response
AI detects an ingredient incorrectly	Users review every item, and uncertain ones are flagged before recipes are generated
AI suggests unsafe or allergenic food	Pass allergies in the prompt, show a reminder, and ask users to check
AI free-tier limit or outage	Show a retry message and keep manual input and saved recipes available
AI returns invalid JSON	Use a response schema, validate the result, and retry once
App Check blocks AI requests in the demo build	Firebase docs list required enforcement from 2 Nov 2026. Use the debug provider in development. For an APK outside Google Play, register its SHA-256, turn off PLAY_RECOGNIZED, and test early
Photos may contain personal content	Gemini free-tier input may be reviewed by people and used to improve Google products. Tell users to photograph only shelves, and do not store photos
Security rules are misconfigured	Test the rules with two accounts before the demo
Hive package maintenance	Stable hive 2.2.3 is about 4 years old. Use the compatible community fork hive_ce if hive causes problems
Short schedule	Keep to the must-have features and leave extras until after the demo

Sources (accessed 2 Nov 2026)

UNEP (2024), Food Waste Index Report 2024: totals p. 64 of PDF, Türkiye data status p. 41, Annex 3 p. 190. Copy: sdg2advocacyhub.org/wp-content/uploads/2024/03/food_waste_index_report_2024.pdf

Nefis Yemek Tarifleri: apps.apple.com/tr/app/nefis-yemek-tarifleri/id947863045, nefisyemektarifleri.com

SnapChef AI: apps.apple.com/us/app/snapchef-ai/id6749183626

Recipez: apps.apple.com/us/app/id6765974915

Samsung Food: sammobile.com/news/samsung-food-update-scan-ingredients-using-phone

Firebase AI Logic: firebase.google.com/docs/ai-logic/get-started, /app-check, /models, /generate-structured-output

Gemini API terms: ai.google.dev/gemini-api/terms. Hive: pub.dev/packages/hive, pub.dev/packages/hive_ce